



5275 - From Dawn to Dusk: Diagnosing Asymmetric Limbs in Exoplanet Transmission Spectra

Cycle: 3, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

We propose an archival study to analyze the publicly available exoplanet transmission spectroscopy data for signatures in the time and spectroscopic domain that indicate the presence of limb asymmetries. Such asymmetries can arise from inhomogeneous temperatures, compositions, and/or aerosol

properties across the terminator and are predicted by 3-D models of exoplanet atmospheres. JWST is the first facility to provide the precision and wavelength coverage needed to unambiguously detect these inhomogeneities for a broad range of planets. Hints of limb asymmetries are already apparent in Cycle 1 data, indicating a wealth of further such discoveries to be made as additional exoplanet data sets become public.

Our innovative data analysis technique involves extracting time-resolved transmission spectra (e.g. separate ingress and egress spectra) and applying novel signal recovery techniques (e.g. catwoman) to obtain a complete picture of the terminator region. This will be accompanied by a forward modeling effort designed to tie our observational results to fundamental predictions for how limb asymmetries manifest as a function of key system parameters such as planetary irradiation.

By measuring limb asymmetries in a total of 19 individual exoplanet atmospheres, we will effectively "spatially resolve" these otherwise unresolved targets. The measurements will provide powerful 3-D diagnostics of aerosols, atmospheric dynamics, and energy budgets that can otherwise be difficult to disentangle in classical transmission spectroscopy. Our analyses will further demonstrate how best to extract information on limb asymmetries from future JWST data sets.

OBSERVING DESCRIPTION